

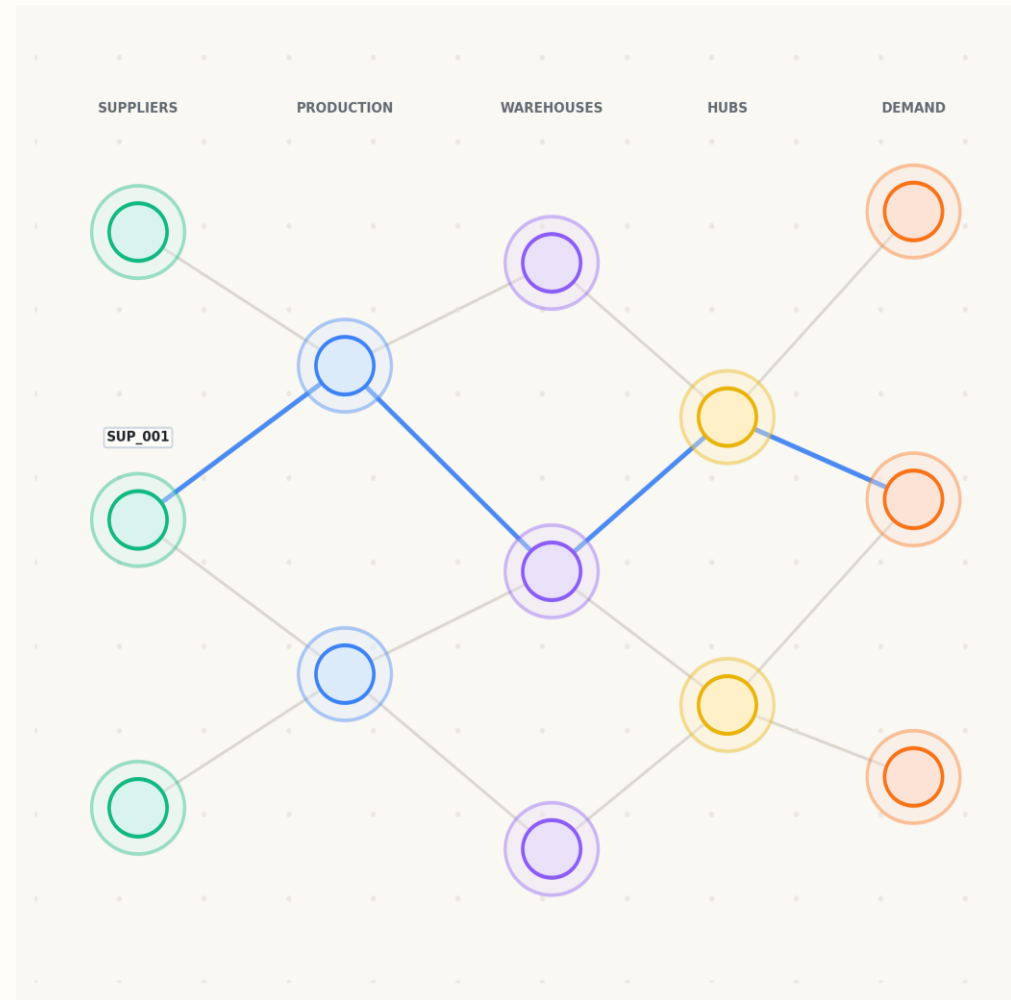
PROJECT REVIEW & TECHNICAL EVALUATION

NEXUS

AI-Powered Supply Chain Intelligence, Risk Prediction & Optimization

MONITOR → PREDICT → IMPACT → SIMULATE → OPTIMIZE → RECOMMEND

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Supply chains don't fail at one point.

A small disruption can cascade across the entire network.

01

FRAGMENTED DATA

Forecasting, inventory, and logistics exist in separate systems, preventing unified end-to-end visibility.

02

DELAYED DECISIONS

Teams often identify the impact only after safety inventory starts running out at distribution hubs.

03

RIGID RESPONSE

Static supplier allocations make it difficult to quickly reroute supply when a primary facility fails.

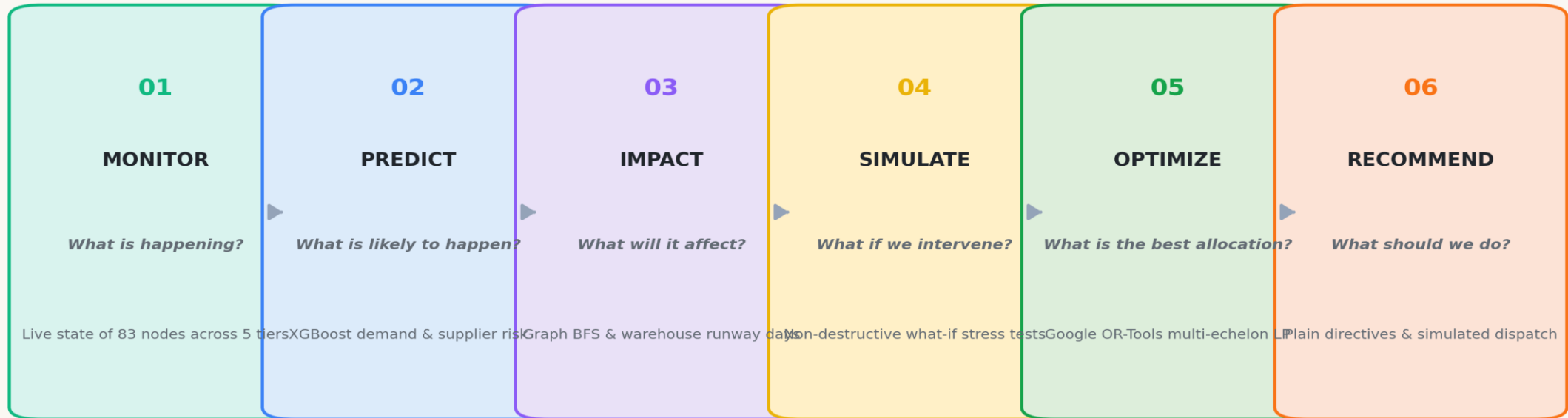
THE DISRUPTION CASCADE:

Supplier Disruption → Inventory Depletion → Production Starvation → Customer Shortages

Without automated intelligence, a regional boiler failure or strike halts downstream retail deliveries within days.

One disruption. One decision loop.

A unified decision intelligence platform closing the gap between prediction and operational action.



← Closed Feedback Loop — Prediction directly drives operational decisions →

Prediction becomes useful only when it leads to a decision.

Built on empirical data + a calibrated digital twin

Combining real-world sales volatility and delivery delays with a multi-echelon spatial topology.

PUBLIC EMPIRICAL DATA

WALMART

421,570

Weekly historical records across 45 stores and 81 departments.

PRIMARY PURPOSE:

DEMAND FORECASTING

PUBLIC EMPIRICAL DATA

DATAACO

35,000+

Representative shipments (180,519 corpus) with authentic delivery delay records.

PRIMARY PURPOSE:

SUPPLIER RISK + BEHAVIOR

CALIBRATED SYNTHETIC

INDIAN TWIN

83 Nodes

160 Corridors, 50,000 Orders, and 500 SKUs mapped across Indian freight corridors.

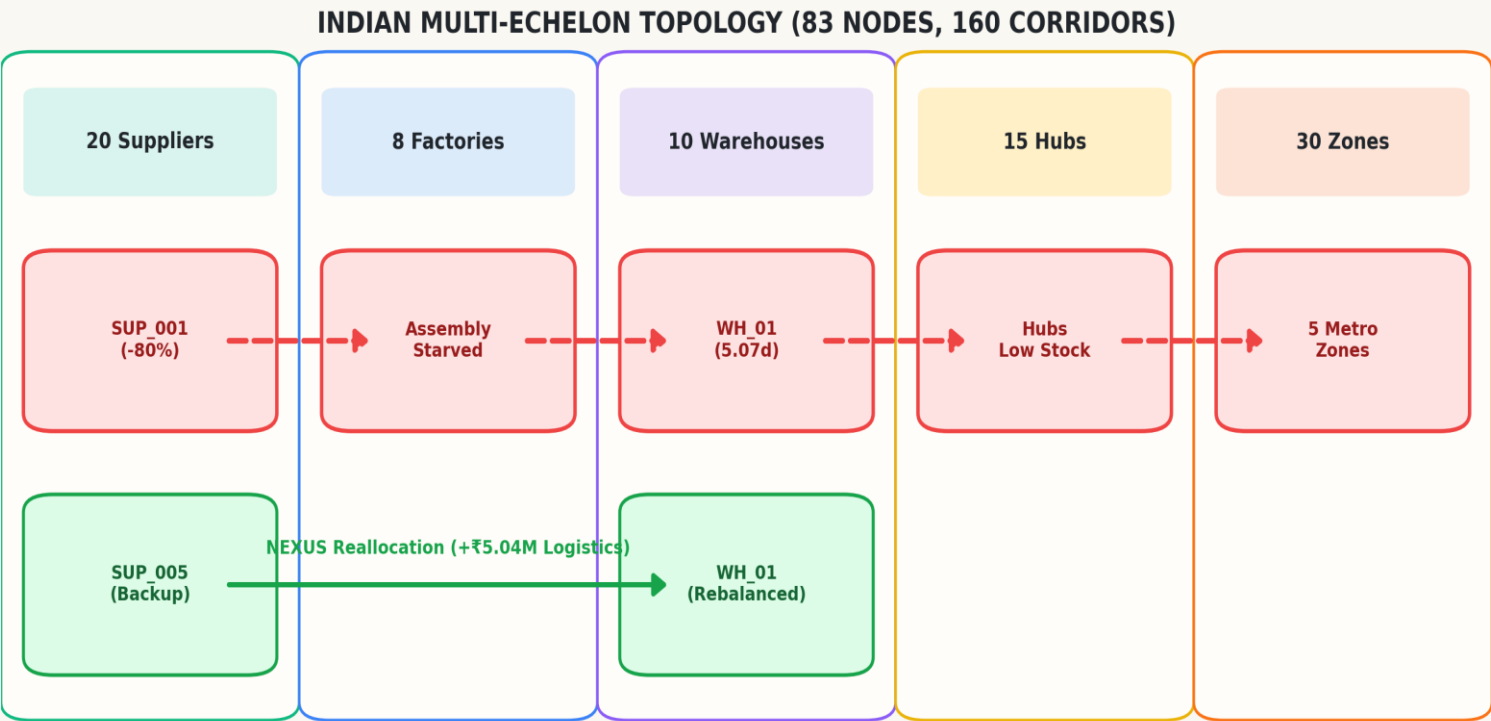
PRIMARY PURPOSE:

SIMULATION & OPTIMIZATION

Academic Integrity: Empirical datasets ground the models; the Indian network is a simulated digital twin.

A supply chain you can see, trace and stress-test.

An 83-node, 160-corridor digital representation across five operational tiers.



Highlighting failure cascade from SUP_001 in Pune to WH_01, alongside optimal secondary vendor rerouting

83

Facility Nodes — Suppliers, plants, warehouses, hubs, zones

160

Freight Corridors — Multimodal highway & rail freight lanes

30

Demand Zones — Tier-1 & Tier-2 consumption centers

₹45.9M

Inventory Value — 500 SKUs tracked across central warehouses

Three models. Three questions.

Machine learning detects the operational signal before failure cascades.

01 — DEMAND FORECASTING

XGBoost Regressor

CORE QUESTION:

"How much will we need?"

OUTPUT SIGNAL:

4-week multi-step demand forecast with seasonal confidence intervals.

02 — SUPPLIER RISK

XGBoost Classifier

CORE QUESTION:

"Who is likely to fail?"

OUTPUT SIGNAL:

Vendor disruption probability based on lead-time volatility and delay history.

03 — ANOMALY DETECTION

Isolation Forest

CORE QUESTION:

"What looks unusual?"

OUTPUT SIGNAL:

Real-time unsupervised detection of sudden demand surges and transit outliers.

ML detects the signal. NEXUS then traces its operational impact.

Does the model actually improve the decision signal?

Empirical out-of-sample evaluation against established industry baselines.

DEMAND FORECASTING (WALMART DATASET)

Model Architecture	RMSE	sMAPE
Naive Persistence	136,255.10	7.20%
4-Week Moving Average	153,446.84	8.42%
XGBoost (NEXUS Model)	96,042.50	4.35%

29.51% RMSE Reduction

XGBoost lowers forecasting error by 29.51% over Naive persistence and 37.41% over 4-Week Moving Average

SUPPLIER RISK (DATACO / 10 UNSEEN VENDORS)

Model Architecture	Precision	Recall	F1
Logistic Regression	0.7531	0.7349	0.7439
XGBoost (NEXUS Model)	0.7442	0.7711	0.7574

77.11% Out-of-Sample Recall

Evaluated via GroupShuffleSplit on 10 completely unseen vendors. High recall ensures early warning of supply failure

From signal → impact → action

Prediction alone does not make decisions; NEXUS connects signals to operational responses.

Signal Detected	Operational Meaning	NEXUS Prescriptive Response
• Demand surge	Safety stock may deplete prematurely	Dynamically increase procurement allocations
• Supplier capacity loss	Critical component shortage risk	Shift sourcing to qualified secondary vendors
• Route delay / blockage	Transit lead-time increases by 2-5 days	Reroute freight via multimodal rail/road
• Low inventory runway	Imminent warehouse stockout (< 7 days)	Trigger inter-warehouse safety stock transfer
• Multi-node shock	Multiple constraints interact across tiers	Execute Google OR-Tools global optimization

NEXUS does not stop at prediction. It connects the prediction to an operational response.

What happens when a critical supplier loses capacity?

Simulating an 80% capacity shock on supplier SUP_001 in Pune.

SHOCK

SUP_001 (Tata AutoComp Pune)
80% Capacity Loss for 10 Days

CASCADE

Western Warehouse (WH_01)
1,420 units stock vs 280 units/day burn rate = 5.07 Days Runway

IMPACT

69,843 Units Potential Shortage
Service level drops to 54.57% without intervention

THE CENTRAL STRATEGIC QUESTION:

"Does the network actually lack capacity?"

+160,750 Units

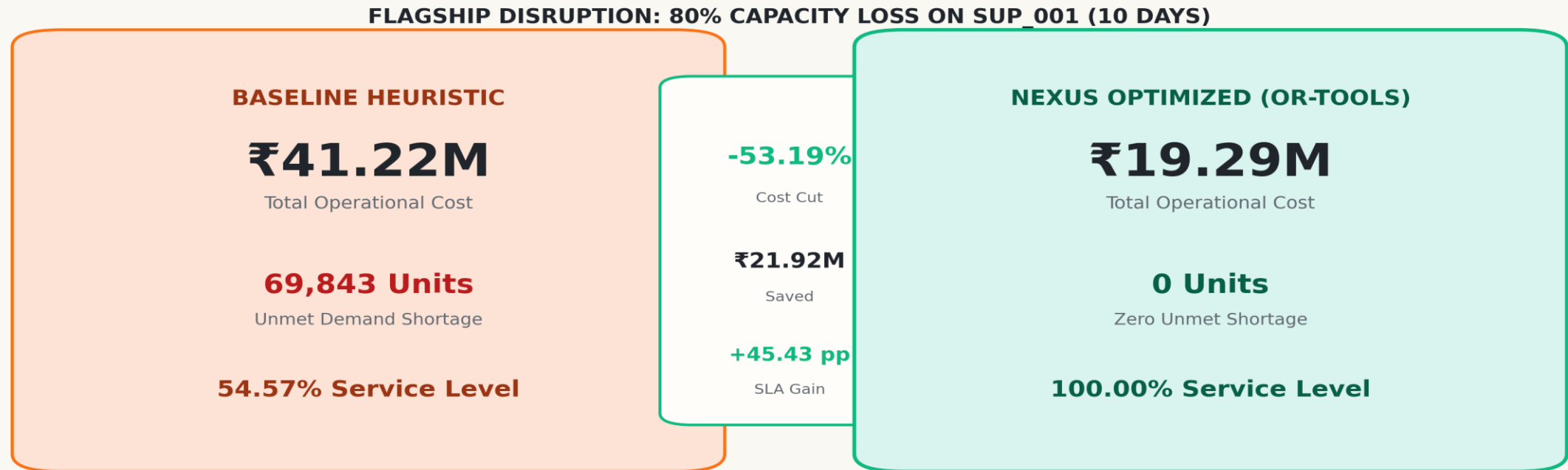
Net Network Capacity Surplus Post-Shock

Total Network Demand = 153,600 units | Remaining Supplier Capacity = 314,350 units.

Key Takeaway: The problem is allocation — not total network capacity.

NEXUS turns the crisis into an allocation decision.

Google OR-Tools multi-echelon linear optimization achieves global cost minimization in 0.0094 seconds.

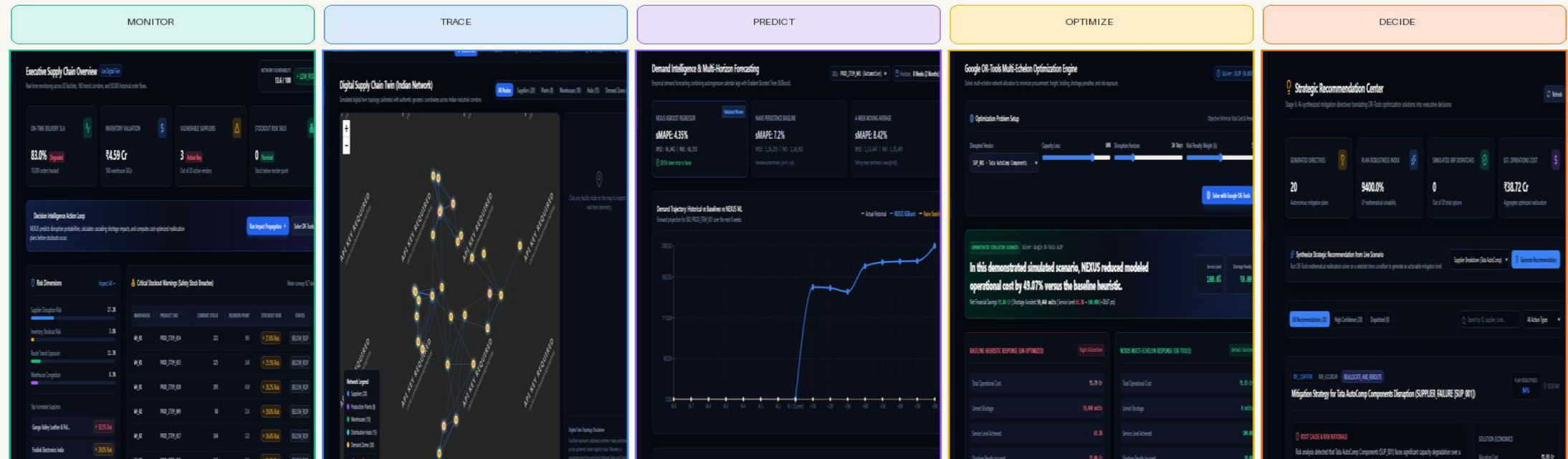


Simulated stress scenario outcome using Google OR-Tools multi-echelon linear program (Solver runtime: 0.0094s)

Simulated stress scenario using Google OR-Tools multi-echelon optimization (Solver runtime: 0.0094s).

From model pipeline to working control room

An interactive enterprise SaaS control room built with React 19, TypeScript, and FastAPI.



TECH STACK: React 19 • TypeScript • FastAPI • Leaflet GIS • Recharts • Google OR-Tools • SQLite / PostgreSQL

Built, tested and verified.

Comprehensive full-stack verification ensuring complete system stability and zero silent failures.

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Backend Tests Passed

Pytest suite (2.66s) covering ML, graph & LP

10 / 10

Frontend Tests Passed

Vitest unit tests for UI and currency formats

0

Build Errors

TypeScript strict compiler & Vite 8.3 bundler

20

REST API Endpoints

FastAPI endpoints verified with HTTP 200 OK

INTEGRATION PIPELINE: React UI → FastAPI → ML / Graph / Simulation → Google OR-Tools → SQLite Persistence

Fixed random seeds (seed=42) • Zero silent mock fallbacks • 100% reproducible execution

Academic Honesty: The platform does not claim live connection to SAP/Oracle or real GPS telematics; ERP dispatch is simulated.

NEXUS in one picture.

Transforming fragmented supply chain data into mathematically verified, actionable decisions.

DATA → UNDERSTAND → PREDICT → TRACE → SIMULATE → OPTIMIZE → DECIDE

"From fragmented supply-chain data to actionable decisions."

NEXUS bridges the gap between predictive AI and operations research, providing supply chain operators with the tools to foresee disruption, trace cascading consequences, and execute optimal mitigation.

Thank You! | Questions & Discussion

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